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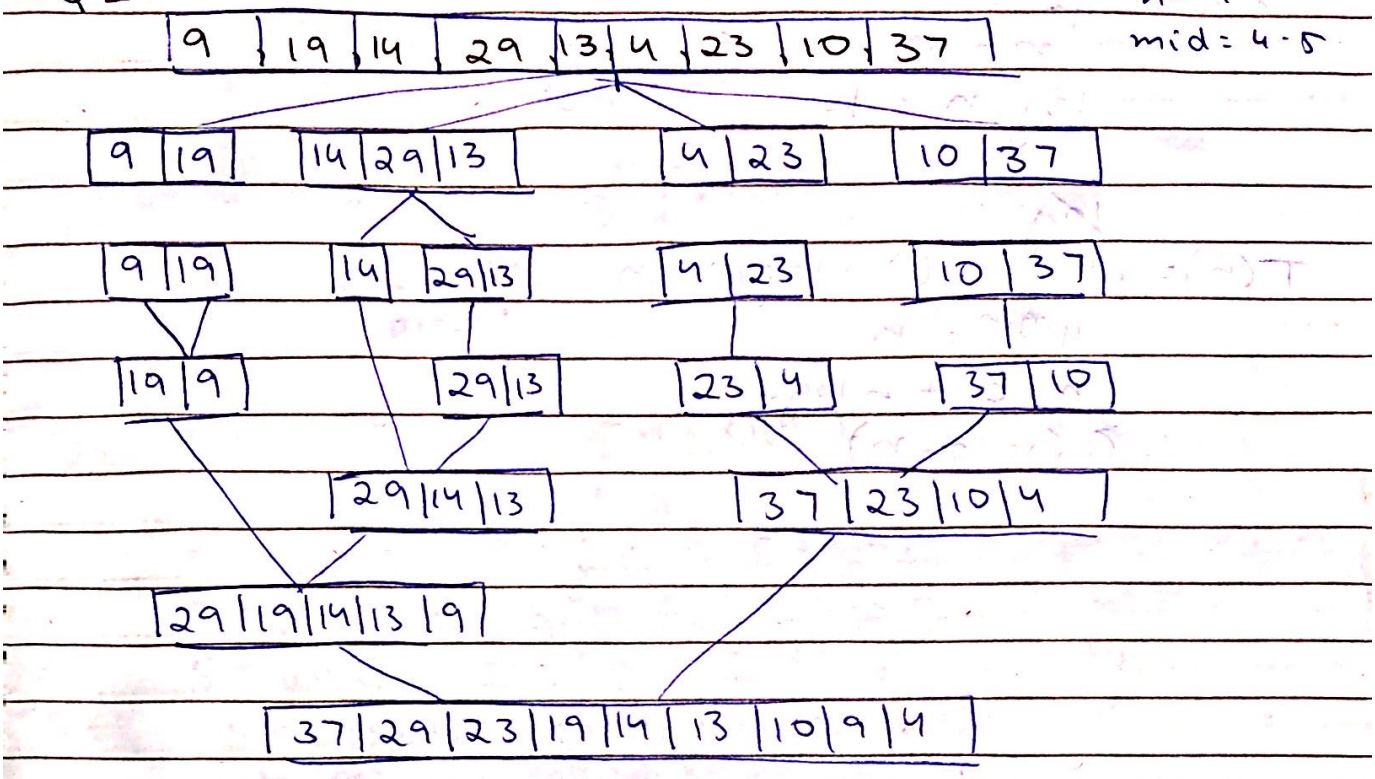
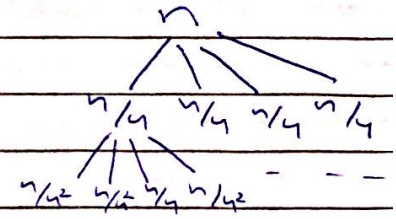
DAA Assignment 1

Q 1.



Time Complexity

$T(n) = 4T(n/4) + n$



$n/4 \Rightarrow T(n/4) = 4T(n/16) + n/4$

$T(n) = 4^2 T(n/4^2) + 2n$

$n/16 \Rightarrow T(n/16) = 4T(n/4^3) + n/4^3$

$T(n) = 4^3 T(n/4^3) + 3n$

after k steps

$T(n) = 4^k T(n/4^k) + kn$

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Q1

$\frac{n}{4^k} = 1 \leftarrow T(1)$

$n = 4^k$

$\log_4 n = k$

$T(n) = 4^{\log_4 n} T(1) + \log_4 n \cdot n$

$\therefore 4^{\log_4 n} = n$

$n \cdot c + n \log_4 n$

$O(n \log n)$

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Q2. $3n^3 + 5n^2 + 25n = \Omega(n^3)$

$f(n) \geq c * g(n)$ $\quad g(n) = n^3$

$3n^3 + 5n^2 + 25n \geq c * n^3$ $\quad c = 1$

$3n^3 + 5n^2 + 25n \geq 1 * n^3$ $\quad n_0 = ?$

$n=1$ $\quad 3(1)^3 + 5(1)^2 + 25(1) \geq 1 * (1)^3$ $\quad n_0 = 1$

$33 \geq 1$ $\quad n \geq n_0$

∴ proved $\quad \Omega(n^3)$ $c = 1$, $n \geq 1$

Q3. $5 \log_2 \log_2 n + 4 \log^2 n = O(?)$

$f(n) \leq c * g(n)$

$5 \log_2 \log_2 n + 4 \log_2^2 n \leq c * g(n)$ $\quad g(n) = \log_2^2 n$

$5 \log_2 \log_2 n + 4 \log_2^2 n \leq c * \log_2^2 n$ $\quad c = 5$

$5 \log_2 \log_2 n + 4 \log_2^2 n \leq 5 * \log_2^2 n$ $\quad n_0 = ?$

$n=1$ $\quad 5 \log_2 \log_2 (1) + 4 \log_2^2 (1) \leq 5 * \log_2^2 (1)$ $\quad * n \geq n_0$

$\log_2(0) \rightarrow$ undefined ×

$n=2$ $\quad 5 \log_2 \log_2 (2) + 4 \log_2^2 (2) \leq 5 * \log_2^2 (2)$ $\quad n_0 = 2$

$0 + 4 \leq 5$

$4 \leq 5$

hence, ~~∴ proved~~

$O(\log_2^2 n)$, $c = 5$, $n \geq 2$

Q4. i. outer loop (i)

for (i = n/2 ; i > 1 ; i /= 6) $\quad$ terminates $i \leq 1$

$-$ assume $i = \frac{n}{2 \times 6^k}$

$\frac{n}{2 \times 6^k} \leq 2 \times 6^k$

$\frac{n}{2} \leq 6^k$

$k \geq \frac{n}{2} : 6^k$

i
$n/2$
$n/2 \times 6$
$n/2 \times 6^2$
$n/2 \times 6^3$
⋮
$n/2 \times 6^k$

$k = \log_6 (n/2)$

$O(\log_c n)$

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Qu i)

mid loop (j)

for (j = 2; j <= n; j *= 4)

j
2
2×4
2×4²
⋮
2×4^K = 2×2^{2K} = 2^{1+2K}

terminates when $j > n$

$2^{1+2K} > n$

$2^{1+2K} = n$

$1 + 2K = \log_2(n)$

∴ ignore constants so

$O(\log_2 n)$

(also shown as $\log_4 n$ if we take 4^K)

inner loop (k)

for (k = 0; k <= j, k *= 3)

terminates when $k > j$

also depends on the j^{th} loop

k
0
0*3
0*3
⋮

runs infinitely

But overall the time complexity (if we ignore the infinite k loop and assume k is starting from 1, it would make k loop complexity as $O(\log_3 n)$ so the time complexity for entire ~~prog~~ algorithm

$i \times j \times k$

$\log n \times \log n \times \log n$

$O(\log^3 n)$

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Q4. ii) Outer loop (i)

for (i = n ; i > 1 ; i = i/2)

i
n
n/2
n/4
⋮
$n/2^k$

terminates $i \leq 1$

$n/2^k \leq 1$

$n \leq 2^k$

$k = \log_2 n$

$O(\log_2 n)$

Inner loop (j)

for (j = 2 ; j <= n ; j += 1)

j
2
3
4
5
⋮
k

terminates when $j > n$

$k > n$

$k = n$

$O(n)$

Overall time complexity

$i \times j$

$\log_2 n \times n$

$O(n \log_2 n)$

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Q5.

$\log\log n^2 \quad \log^2 n \quad n^{-2} \quad \log(2^{\log(n^2)}) \quad n^{1/2} \quad n^2$

Simplify for easier calculation

~~Apply log~~

$\log(2\log n)$	$(\log n)^2$	$\frac{1}{n^2}$	$\log(2^{2\log n})$	$n^{1/2}$	n^2
$\log 2 + \log\log n$	$\log n \log n$		$\log(2^{\log n})^2$	$\sqrt{n}$	
$1 + \log\log n$			$\log(n^{\log 2})^2$	$n^{1/2}$	
~~$\log\log n$~~			$\log(n^2)^2$		
			$2\log n$		

Mathematically

n	$1+\log\log n$	$\log n \log n$	$\frac{1}{n^2}$	$2\log n$	$n^{1/2}$	n^2
2^1	$1+0=1$	1	$1/4$	$2(1)=2$	$\sqrt{2}$	4
2^4	$1+2=3$	16	$1/256$	$2(4)=8$	$=8$	256
2^8	$1+3=4$	64	$1/65536$	$2(8)=16$	$=16$	65536
2^{128}	$1+7=8$	16384	$\frac{1}{2^{128}}$	$2(128)=256$	1.84×10^{19}	$(2^{128})^2$
	(2)	(4)	(1)	(3)	(5)	(6)

We confirm values on larger n values so non-decreasing / increasing order is:

$$n^{-2} < \log\log n^2 < \log(2^{\log(n^2)}) < \log^2 n < n^{1/2} < n^2$$

Q6. We measure efficiency of an algorithm by its speed. we measure speed in terms of the operations performed and the no. of inputs.

We use Asymptotic Notation to determine the speed or time complexity : Big O, Big Ω, Big Θ

~~When f(n)~~ Big O is ~~if~~ $f(n) \in O(g(n))$ if there are positive constants c & $n_0 > 0$ such that $f(n) \leq c * g(n)$ for all $n \geq n_0$.

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Q6. Big Ω is $f(n) \in \Omega(g(n))$ if there exists positive constants c & n_0 such that $f(n) \geq c * g(n)$ for all $n \geq n_0$.

Big Θ is $f(n) \in \Theta(g(n))$ if there exists positive constants c_1, c_2, & n_0 such that

$c_1 \times g(n) \leq f(n) \leq c_2 * g(n)$ for all $n \geq n_0$

- Algorithms : we will use integer functions & inputs
- n is real variable, n is integer variable

But the definitions of $O(n)$, $\Omega(n)$ & $\Theta(n)$ is same for n

Q7. i) $n = w\sqrt{n}$

$f(n) > c \cdot g(n)$

$n > \frac{1}{} \cdot \sqrt{n}$

$n > \sqrt{n}$

$n = 1$ $\quad 1 > \sqrt{1}$ $\quad \times$

$n = 2$ $\quad 2^1 > \sqrt{2}$ $\quad \checkmark$

$g(n) = \sqrt{n}$

$c = 1$

$n_0 = ?$

$n \geq 2$

$n = w\sqrt{n}$ is True for $c = 1$ & $n \geq 2$

ii) $f(n) = O(g(n))$ then $g(n) = \Omega(f(n))$

Assume $f(n) = n$ $\quad g(n) = n^2$

$f(n) = O(g(n))$

$n = O(n^2)$

$f(n) \leq c * g(n)$

$c = 1$ $\quad n \leq \frac{1}{} * n^2$

$n \leq n^2 \rightarrow O(n^2)$

$n_0 = ?$ $\quad 1 \leq 1$

$n \geq 1$

$g(n) = \Omega(f(n))$

$n^2 \quad c = 1 = \Omega(n)$

$g(n) \geq c * f(n)$

$n^2 \geq \frac{1}{} * n \quad c = 1$

$n^2 \geq n \rightarrow \Omega(n)$

$1 \geq 1 \quad n \geq 1$

So True

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Q7

iii) $f(n) = O(f(n)^2)$

Assume $f(n) = \frac{1}{n}$ so presumes $O((\frac{1}{n})^2)$

$f(n) \leq c * g(n)$

$\frac{1}{n} \leq c \cdot \frac{1}{n^2}$

$\frac{1}{n} \leq c \cdot \frac{1}{n \times n}$

$1 \leq c \cdot \frac{1}{n}$

∴ constant is not less than a decreasing function

so False

iv) if $O(f(n))$ & $\Omega(f(n))$ then $\Theta(f(n))$

Assume $f(n) = 2n+3$ $g(n) = n$

$O(f(n))$

$f(n) \leq c \cdot g(n)$

$2n+3 \leq \underline{5}\, n$ $c=5$

$2(1)+3 \leq 5(1)$ $n_0 = ?$, $n=1$

$5 \leq 5$

$O(n)$

$\Omega(f(n))$

$f(n) \geq c \cdot g(n)$

$2n+3 \geq c \cdot n$ $c=1$

$2n+3 \geq \underline{1} \cdot n$ $n_0 = 1$

$2(1)+3 \geq 1(1)$ $n=1$

$5 \geq 1$

$\Omega(n)$

$\Theta(f(n))$

$c_1 \cdot g(n) \leq f(n) \leq c_2 \cdot g(n)$

$c_1 \cdot n \leq 2n+3 \leq c_2 \cdot n$

$1 \cdot n \leq 2n+3 \leq 5n$

$n \leq 2n+3 \leq 5n$

$n_0 = 1$ $1 \leq 5 \leq 5$ → $\Theta(n)$

so True

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Q8. Polynomial Time

1. Input n no. of process — O(1)
2. Input time for each process — O(1), burst_time[i]
3. For every p[i] ; time_left [i] = burst_time [i] - O(1)
4. for (i = 0 to n-1) { — } O(n²)
   for (j = 0 to n-i-1) {
   if burst_time [j] > burst_time [j+1] {
   swap [burst_time [j] , burst_time [j+1]
   swap (process [j] , p [j+1]);
   } } }
5. Initialize sum = 0, turnaroundtime = 0 — O(1)
6. for (i = 0 to n) { — O(n)
   sum = sum + burst_time [i]
   turnaroundtime = turnaroundtime + sum
   }
7. Return total turnaroundtime → O(n²)

Logarithmic Time

* Input process [n]
* Function Mergesort (Process)
  mid = Process.length /2 ;
  left [] = Process [0 -> mid]
  right [] = Process [mid -> n]
  merge (left)
  merge (right)

function merge (left, right) {
  initialize i = 0 , j = 0, k = 0
  while (i < left.length && j < right.length)
  if (left [i] < right [j])
  Process [k++] = left [i++]

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Q8. else

Process [k++] = right [j++]

while i < left.length

Process [k++] = left [i++]

while j < right.length

Process [k++] = right [j++]

Sum = 0, turnaround time : 0

For each burst time in Processes

Sum += burst_time

turnaroundtime += sum

Return turnaround time

O(nlogn)